

Divisibility and the Division Algorithm

- **Exercise 1**

Let $d \in \mathbb{N}$ and suppose that the division of $n \in \mathbb{Z}$ by d gives quotient q and remainder r ,

$$n = dq + r \quad \text{with} \quad 0 \leq r < d.$$

Determine, in terms of p, q and k , the quotient and the remainder obtained when $n + kd$ is divided by d .

We need

$$n + kd = (\quad)d + ? \quad \text{with} \quad 0 \leq r < d.$$

- **Solution 1.1**

$$n + kd = dq + r + kd = d(q + k) + r.$$

Then the quotient is $(q + k)$ and the remainder is r .

- **Exercise 2**

Let $a_1, a_2, \dots, a_n \in \mathbb{N}$ that satisfy $a_1 \mid a_2, a_2 \mid a_3, \dots, a_{n-1} \mid a_n$. Suppose that $a_1 < a_2 < \dots < a_{n-1} < a_n$. Prove that $\forall n \in \mathbb{N}, a_n \geq 2^{n-1}a_1$.

- **Proof 2.1**

What happens when $n = 1$: $a_1 \geq 2^{1-1}a_1$.

What happens when $n = 2$: $\exists k \in \mathbb{N}, a_2 = ka_1$. And $k \geq 2$ since $a_1 < a_2$. Then $a_2 \geq 2^{2-1}a_1$.

What happens when $n = 3$: $\exists k \in \mathbb{N}, a_3 = ka_2$. And $k \geq 2$ since $a_2 < a_3$. Then $a_3 \geq 2a_2 \geq 2 \cdot 2^{2-1}a_1 = 2^2a_1 \implies a_3 \geq 2^{3-1}a_1$. From this, we can prove the statement by induction.

1. Base step

We already proved that the result is true for $n = 1$.

2. Inductive step

Suppose that the result is true for n , that is,

$$a_n \geq 2^{n-1}a_1.$$

Now consider a_{n+1} . We have $a_n \mid a_{n+1}$ and $a_n < a_{n+1}$. By definition, $\exists k \in \mathbb{N}$ and $k \geq 2$ such that $a_{n+1} = ka_n$. Then we have

$$a_{n+1} \geq 2a_n \geq 2 \cdot (2^{n-1}a_1) = 2^n a_1.$$

Then the result is true for $n + 1$ and therefore

$$a_n \geq 2^{n-1}a_1$$

is true for every $n \in \mathbb{N}$.

- **Exercise 3**

Determine all integers d such that $d \mid 84$ and $d \mid 126$.

- **Solution 3.1**

We want to prove that the set of all integers d such that $d \mid 84$ and $d \mid 126$ is exactly $\text{Div}(42)$.

If d is one of those that we want to find, i.e., $d \mid 84 \wedge d \mid 126$, then d divides any integer combination of them. As $42 = 126 - 84$ is an integer combination of 84 and 126, then $d \mid 42$.

Conversely, suppose $k \in \text{Div}(42)$, then

$$\begin{aligned} k \mid 42 \wedge 42 \mid 84 &\implies k \mid 84, \\ k \mid 42 \wedge 42 \mid 126 &\implies k \mid 126. \end{aligned}$$

Therefore, the required integers are exactly the elements of $\text{Div}(42)$.

• **Exercise 4**

Let $d, e \in \mathbb{N}$. First, divide $n \in \mathbb{Z}$ by d , that is, $n = dq + r$, $0 \leq r < d$. Next, divide q by e , that is, $q = es + t$, $0 \leq t < e$.

Prove that when n is divided directly by de , the quotient is s and the remainder is $dt + r$.

◦ **Proof 4.1**

$$\begin{aligned} n &= dq + r \\ &= d(es + t) + r \\ &= des + dt + r \\ &= (de)s + (dt + r). \end{aligned}$$

Then we want to prove $de > dt + r \geq 0$. Since $t \leq e - 1$, we have:

$$dt + r < d(e - 1) + d = de.$$

Also, since $d \in \mathbb{N}$, $t \geq 0$, and $r \geq 0$, we have $dt + r \geq 0$. Thus, $0 \leq dt + r < de$, satisfying the remainder condition.

• **Exercise 5**

Let $a, b \in \mathbb{Z}$. Suppose $a \mid b \wedge a \mid (b + 1)$. Prove that $a = 1 \vee a = -1$.

◦ **Proof 5.1**

Suppose $a \mid b \wedge a \mid (b + 1)$. Then a divides any integer combination of b and $b + 1$. Since

$$\begin{aligned} 1 &= (1)(b + 1) + (-1)b \implies a \mid 1 \\ &\implies \exists k \in \mathbb{Z}, 1 = a \cdot k \\ &\implies a, k \in \{1, -1\} \\ &\implies a = 1 \vee a = -1. \end{aligned}$$

• **Exercise 6**

For $a, b \in \mathbb{Z}$, prove that $\text{Div}(a) = \text{Div}(b) \iff a = b \vee a = -b$.

◦ **Proof 6.1**

($\implies$) Suppose $\text{Div}(a) = \text{Div}(b)$.

As $a \in \text{Div}(a) \implies a \in \text{Div}(b) \implies a \mid b \implies \exists k \in \mathbb{Z}, b = ak$.

As $b \in \text{Div}(b) \implies b \in \text{Div}(a) \implies b \mid a \implies \exists q \in \mathbb{Z}, a = bq$.

Then $a = bq = akq$.

If $a \neq 0$, then $kq = 1 \implies k = 1 \vee k = -1 \implies b = a \vee b = -a$.

If $a = 0$, then $\text{Div}(0) = \mathbb{Z} = \text{Div}(b) \implies 0 \in \mathbb{Z} \wedge 0 \mid b \implies b = 0$. Thus $b = a$.

($\impliedby$) Suppose $a = b \vee a = -b$.

If $a = b$, then $\text{Div}(a) = \text{Div}(b)$.

If $a = -b$, suppose $d \in \text{Div}(a) \implies \exists k \in \mathbb{Z}, a = dk$. Then $b = -a = -dk = d(-k)$ for some $-k \in \mathbb{Z}$. Then $d \mid b \implies d \in \text{Div}(b) \implies \text{Div}(a) \subseteq \text{Div}(b)$. In a similar way, we can prove that $\text{Div}(b) \subseteq \text{Div}(a)$. Hence $\text{Div}(a) = \text{Div}(b)$.

• **Exercise 7**

Find all integers $d \neq 0$ for which the remainder obtained when 47 is divided by d is 5.
For every possible value of d , determine the corresponding quotient.

◦ **Solution 7.1**

By the definition of the division algorithm, we have:

$$47 = dq + 5 \quad \text{with} \quad 5 < |d|.$$

This implies $dq = 42$, which means $d \in \text{Div}(42)$ and $|d| > 5$.

The set of all divisors of 42 is $\text{Div}(42) = \{\pm 1, \pm 2, \pm 3, \pm 6, \pm 7, \pm 14, \pm 21, \pm 42\}$.

Filtering for $|d| > 5$, we find the possible values for d :

$$\text{Ans} = \{\pm 6, \pm 7, \pm 14, \pm 21, \pm 42\}.$$

The corresponding quotients $q = \frac{42}{d}$ for each possible value of d are given below:

- If $d = 6$, then $q = 7$.
- If $d = -6$, then $q = -7$.
- If $d = 7$, then $q = 6$.
- If $d = -7$, then $q = -6$.
- If $d = 14$, then $q = 3$.
- If $d = -14$, then $q = -3$.
- If $d = 21$, then $q = 2$.
- If $d = -21$, then $q = -2$.
- If $d = 42$, then $q = 1$.
- If $d = -42$, then $q = -1$.

• **Exercise 8**

Prove that $\forall n \in \mathbb{N}, (a - b) \mid (a^n - b^n)$.

◦ **Proof 8.1**

By induction.

1. Base step

The result is true for $n = 1$ because $(a - b) \mid (a^1 - b^1)$.

2. Inductive step

Suppose that the result is true for n , that is,

$$(a - b) \mid (a^n - b^n).$$

By definition, $\exists k \in \mathbb{Z}, k(a - b) = (a^n - b^n)$. We want to prove that it is true for $n + 1$.

We have

$$\begin{aligned} a^{n+1} - b^{n+1} &= a \cdot a^n - b \cdot b^n \\ &= a \cdot a^n - (b - a + a) \cdot b^n \\ &= a \cdot (a^n - b^n) + (a - b) \cdot b^n \\ &= a \cdot (k(a - b)) + (a - b) \cdot b^n \\ &= (a - b)(ka + b^n). \end{aligned}$$

Since $ka + b^n \in \mathbb{Z}$, $(a - b) \mid (a^{n+1} - b^{n+1})$. So the result is true for $n + 1$ and therefore the statement is true.